- I originally started this project with the intent of curating a dataset using four relatively understudied succulents in terms of western pharmacology: Sanseviera Triasfaciata, Sansevieria Zeylanica, Bryophyllum Pinnatum, and Zyglophyllum Coccnieum.

- I went through different papers for each plant, scrapping up the names of each metabolite for each plant from different literature. I decided to create a organizational file tree whereby the this top directory would be the plant, then the chemical family names, then the individual listings of each secondary metabolite.

- I ran into a problem whereby most of these plants had very little metabolic literature on them, but Bryophyllum Pinnatum had a lot of information for it. I found that at least 85 different metabolites had been identified, while for some of the other plants I found there were maybe 10 or so that were mapped. Given that the dataset would be extremely unbalanced and that it was really difficult to organize all my existing collections of metabolites for each plant without getting things extremely confused by mixing up names of metabolites and the chemical families, I decided to stick with Bryophyllum Pinnatum.

- In order to map out all recorded chemicals of Kalanchoe Pinnata for my model, I scrapped my old file stack and started by extracting all registered secondary metabolites from experimental research papers, creating a .JSON file with the names of the metabolites, their pubchem ID's, and their SMILES strings. This way I could use the Chem.AllChem and Chem.draw libraries to created 3D visualizes of these metabolites and train my models on them.

- The problem here was that the pubchem ID's for these chemicals were actually just for the general form of the chemicals. This means that I wasn't actually going to visual the chemicals as how they were to appear in the plant, but rather how they would appear is a vacuum with no other context given to them. I tried to get the code to extract any information from my curated list of papers on how the molecules were formed in terms of the anatomy of the plant from which they were found, but I still got the same results.

- I decided to start over from scratch and do the same thing of extracting the names, pubchem ID's, and SMILES strings, but this time I added a layer where I record where exactly within the plant anatomy (plant tissue, leaf buds, stem, leaves, etc.) were these metabolites located. I then would use that to create a multi-modal dataset with pictures of these leaves so that I could create my dataset and use it to train the model. I realized this would be an issue, however, when I realized all of my images were macrolevel pictures of the plant rather than microscopic images of the plant tissue like I needed.

- To make this still work, I decided to delete all the images I had for each folder and went out looking for available images of plant tissue. I was only able to find three or so images, mainly of the rib and some that were just clusters of cells that had undergone multiple rounds of manipulation for research on the plant's reaction to environmental constraint and it's medicinal properties. I had wanted to duplicate the images I had found

so that I would have enough for my model, but I realized that would comprise the integrity of my model by teaching it only how to classify one tissue image and the metabolites that can be found on their rather than the anatomy of the whole and all recorded metabolites.

- I decided to scrap that and looked for other ways to capture these metabolites. I decided that rather than doing the multi-modal dataset with images, I could make graphical simulations of the plants tissue features and train my model using those features instead since I didn't have any pictures of the plants tissue that would work for my model. That was the problem. Because I didn't have actual tissue images of the plant, there was no way I could know whether the simulations were true to the plants anatomy, or if they were complete hallucinations not even kind of resembling the model's real features.

- Given that, I decided doing that would be a very bad idea, so I decided to scrap the idea all together, and I turned my attention to metabolomics data. On the first try of this, I was able to find a true metabolomics dataset curated specifically for Bryophyllum Pinnatum. I thought it was perfect. I instantiated the model, generated the t-SNE graph for its predictions, and then went ahead to grab more data to train my model with. I realized that as Bryophyllum Pinnatum wasn't a model organism, there were very few available metabolomics datasets that perfectly mapped out the location of the secondary metabolites in relation to the plant. Most datasets stop at genomics data.

- Because of this, I decided to scrap the entire thing again. I decided to stop this loop by taking some time to read more about the plant's anatomy, its medicinal properties, and how these metabolites interact within the plants anatomy under environmental changes and other conditions.

- After going through these papers, I came up with the idea to base the network topology of my HeteroGNN based on the Crassulacean Acid Metabolism pathway. I extracted a few diagrams of the pathway through the experimental papers I read, and I decided that I would use that as the basis of my topology. I was able to find mass spectrometry and chromatographic data that I could use to map out the peaks of the metabolites in relation to where the CAM topology layout, but I realized that by merely dropping in a bunch of unorganized metabolomic datasets into my model, the integrity of my model would be severely compromised.

- So I decided that rather than do that I'd keep the CAM network topology structure, and what I would do is learn more how analytical chemistry actually works so that I can use some of that information to define the topology. I figured this would be the best way to go about it because each molecule in a plant would have different retention times and retention factors based on the method in which the chromatography is being executed. I will also be combining this with the Blasius differential equations for modeling the oscillatory rhythm of the stomata's opening and closing pattern which influences a late concentration and CO2 processing. Rather than plotting the equations using arbitrary

values that may or may not actually reflect the true nature of the connections between all these equations, I decided that it would be better instead to have my code to graph the connections using jacobian matrices. This would allow me to see where the model points naturally intersect along their graphs rather than forcing them into some arbitrary setup.

- In my initial setup I realized I didn't actually execute the derivations and just straight up plugged in the equations. This would be problematic because it wouldn't allow me to map the diurnal gradients that influence the nature of the oscillation. To fix this I went back to the equations and found the necessary partial derivatives for each variable in each equation. I then put it in vector form as my first set of vectors.

- This solved the first problem with my setup, but it left me with another hole in my structure. How would I be able to link the analytical chemistry equations and variables of retention time and retention factor to the oscillation models? I then thought back to Professor Leonard's lesion of directional derivatives, and I thought about how he showed that the direction derivative vector was essentially the combination of the gradient vector and the unit vector derived from a specifically given vector. I realized that what I could do was I could use the analytical chemistry equation as vector components in the x and y directions (there was no z component with this setup). I did that, then I found the magnitude those components would make. This served as my unit vector, and the oscillation vector I got would serve as my gradient vectors.

- Now the goal was to model this as a natural slope because these equations show the interlocking nature of compounding variables that greatly influence the rate and the structural nature of metabolic production. So as to then model this whole interaction as a slope, I thought again back to Professor Leonard's video. He said that to get the directional derivative from this point you would dot product the unit and gradient vectors together. So that's what I did, and that slope is to model the interlocking nature of metabolite structure as a product of it's environment.

- So with that issue fixed, I realized there was another problem with my system. The CAM graph scheme I was using to ground my topology was purely diagrammatic. This would make it extremely difficult to interlock my slope derivations with the CAM network. So I decided to scratch that and thought about how I could connect the two together. I looked through the data supplements I had garnered from the papers I read, and after some time of trial and error, I landed on a qualitative topological scheme that related qualitative connection between things like temperature, water, carbon dioxide, photoperiod, metabolic fluctuation, salinity, transpiration, etc. together. I figured this would be perfect because it did two very important things. First, because of how it used a node and edge structure to map out the flow of these variables I could use it to define the structure of my graph. Second, it enabled me to uncover so many other hidden commodities I would have otherwise forgotten to account for when building my model, which is especially important because Bryophyllum Pinnatum, like all other Crassulacae plants exhibiting CAM metabolism, utilizes a diurnal cycle in which the conditions of photosynthesis shift

repeatedly depending on the conditions the plant is faced with in the moment. It would enable me to further ground my topology in environmental reality rather than just in an idealized system that doesn't exist.

- In order to make use of this topology, I thought back to Professor Strang's lecture on Graph Matrices, specifically as to how he set up the matrix which he derived from the node and edge connections. I replicated that structure, but because many of the connections had bidirectional flow modeled, I decided that I would turn my graph into a 2D rather than 1D graph to account for those as well without compromising the mathematical integrity of my model.

- So I came across the problem of making sure the matrices were related back to the gradient magnitudes I found earlier from the oscillation models and the analytical chemistry equations. Then I thought about what happens when you multiply a scalar to a vector. The scalar value would serve as the component defining the vector's trajectory. Since a matrix is essentially the compression of multiple vectors in a system to span a particular dimensional subspace, I can use my dot product scale to differ the trajectory of my topological matrices, thereby fully connecting both into a cohesive system.

- In order that the data points of my matrices properly connect back to the constraints of the plant, I decided it would be best to look back at the exact constraints the topology was linking together. I decided to hunt for the properly mathematical analytics and chemical systems to define these systems by. This is what I would use to define my connections when I go to instantiate my topology.

- At first I had wanted to use the K. Pinnata findings on the correlation to phtonihbithion distinctions between sun and shaded leaves and their quantum CO2 yields. They had graphs and data explicitly mapping these two together and I believed it would be perfect since I would extract the graphical structure, export the coordinates to Desmos, and I would get the model of best fit to represent the dynamic shift in variables.

- I went to render the first graph. None of the features were accurately rendering the graphical coordinates with the proper scientific context. Some came close but they produced an unnatural continuous curve when the actual scientific graph was extremely jagged with splits in between multiple periods. The points would map back to their own space and not that which was used in the table.

- I then tried to feed the raw data to these graph-rendering sites thinking I would get something because I was just rendering information, but many of the same issues occurred and I felt like there was probably a better way I could go about this.

- I decided to no longer use those data points specifically, and I decided to pivot to an article that, while not necessarily related to B. Pinnatum, discussed the combining variables of Crassulacean Acid Metabolism a lot like how the Blausius Models did. They provided concrete equations to define their setup and marked what each variable was supposed to show. I decided that doing this would preserve my model's integrity by keeping it rooted in scientific truth rather than relative curve matching. The plan is to continue this process of finding and manipulating equations to act as a weight for the corresponding point on the matrix until they are all done. They will be integrated one by one as part of a larger multivariable slope topology. This process will take a while given as they are sourced from different scientific papers and experiments, but doing it this way would preserve the physical and biological grounding of the topology.

- For some like Photorespiration, I was able to solve for the necessary variable, and that will be the weight that acts on one of the matrix points.

- Then I solved and got a multivariable differential function that was perfect for Stomatal Conductance/Transpiration. I couldn't just solve for a particular variable because all components played a major role in the system of transpiration and it wasn't just delegated to one variable. This also defined another point with my 2D matrix.

- For Ion Balance, I found an article that deals the with electrophoresic scaling of malate ^2- in terms of the tonoplascts of Ca+, K+, and H+ in particular because of their heavy role in the production NADP molecules. This lead me to find the paper that talks about this in mathematical form, and it was the Nernst Equations for electric transport equilibrium modeled as ionic transport gradient flow. This equation, when plugged in for Malate^2-, H^+, and Cl^- will be part of the weight for ionic transport.

- For C/N signaling, I found an equation for nitrate optimization under biomass and photon constraints that came from the Michaels-Menten Equations, and I loved the amicus nitrate concentration. I then found another series of ODEs that model the role of enzyme srnk1 and the other enzymes/substrates that influence it and are influenced by this enzyme. This is important because srnk1 is essentially the molecule behind carbon regulation through carbohydrate metabolism. I modeled the unified differential slope from the gradients of these two equations to show how each molecular interaction is playing an active role in the cycle of carbon. With these two things, I set up the Cfix/Nfix ration for photosynthesis, which will serve as the C/N signaling weight.

- I used the portion of the Blausius Equations specifically with respect to vascular pressure to create the mathematical weight for modeling osmotic reactions within the plants cellular components.

- For temperature, I found six different equations that have to be accounted for in creating that specific weight for my topology. For the 1st equation, the flux setup, in the paper I read I saw that the original researcher separated the methodology into three different parts. This enabled me to perform multiple substitutions to get a master function for that that tied everything together. This served as my first function. I saw that the researcher had another equation set exactly like the same way as the first framework. I did the same thing, which is that I got the master equation by substituting different parts of the top equation with nestles variables defining each part. So this ultimately left me with two equations up to this point. With these first two equations I cross-producted them to get an orthogonal vector out of them. I decided this would be the best route to take so as to ensure that both conditions and configurations from each function are being directly compounded in the final vector. I then turned this into a unit vector because I was going to combine it with another vector. The vector I combined it with was the final function that I got using the exact same steps I used to get the first and second master equations. I took partial derivatives from this equation to get a gradient vector to act as another major compounding force on the overall weight. I dot-producted my final orthogonal unit vector function with the gradient to get the directional derivative that will serve as my entire temperature weight.

- For modeling membrane fluctuation patterns in regards to carbon assimilation, I had originally wanted to do a dot product between two of my earliest gradient vectors for stomatal conductance and then I would take that and dot product it with my Nernst equation for ion transport. But there was one problem, and this problem was that these processes aren't separate but are actually connected in the same process. Not only that but I realized that I would have to track the amount collected over the nighttime period and then the amount that was left over the daytime period. I realized that I would need two different sets of double integrals to model this setup. The inside equation for the integral was my Nernst equation because the ION transport is what is controlling the membrane cycling, and I set the bounds of integration to the Carbon assimilation slope expression from one given hour to the next given hour because these are the parameters that control the Bentsen fluctuation accumulation. The second integral was an additional finite series starting from the first hour to the 12th hour day or night because the accumulation within a single integration sequence would vary depending on the already present carbon accumulation within the vacuole in relation to the hour being tracked within that single run, so while there would likely being a pattern to the rate of carbon accumulation in relation to the ion transport cycle the actual values would likely not remain equal, so adding the additional sum in there would counter-balance that.

- For the night flux balance, I started the t (standing for time) at one, and I started the lower bound of my integration with the t hour minus one and the upper bound of integration at the t hour. I set it this way so that I can track the amount of carbon assimilation at the 0th hour easily without having to set n equal to 0 for my summation.

- For my day influx I just reversed the positions of the integration bounds still tracking my summation to the 12th hour. I did this because with each hour the membrane is releasing some of the carbon and with each hour tracking backwards the carbon would've been less and less.

- For metabolic induction, I viewed this weight as another combination of factors. I thought of it less as what causes it and more about how the membrane fluctuation cycle functions within the compounding effects of soil conductance and the factors that influence that. I solved for the decreasing rate of water conductance in water stressed environments, getting the slope of the resulting function. Then I decided to multiply that slope directly with the variable to create a self-reinforcing expression. For the root index when the roots are water saturated, I did thing but this time I didn't map out the slope but instead just multiplied the individual vector without its slope. I did this because there wasn't an extra condition that needed to be accounted for relating directly to it, as all necessary components were parlayed within the expression. I chose the water constraints for this because one key condition for CAM metabolism is vacuole capacity, and the vacuole is what holds the most water in a plant cell, and the exchange of water and CO_2 is what drives photosynthesis.